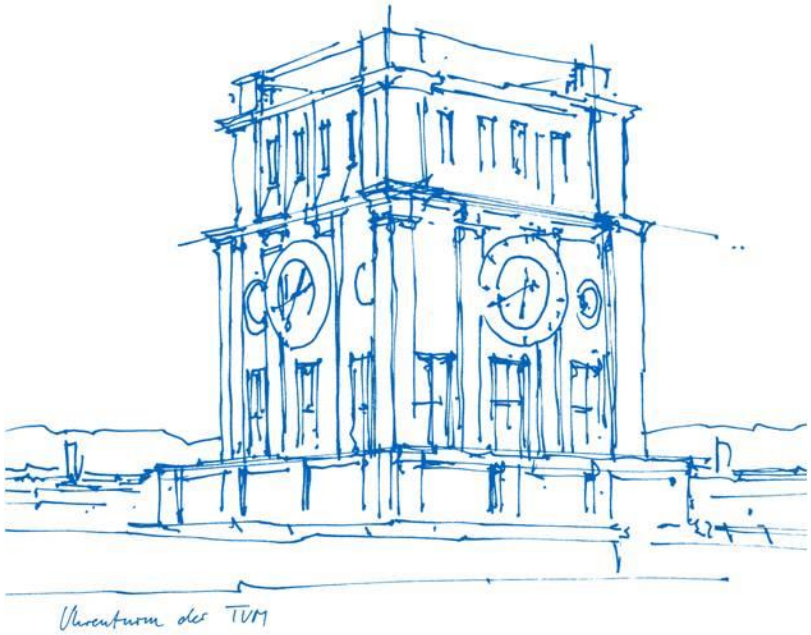
Chair of Computational Mathematics
TUM School of Engineering and Design
Technical University of Munich

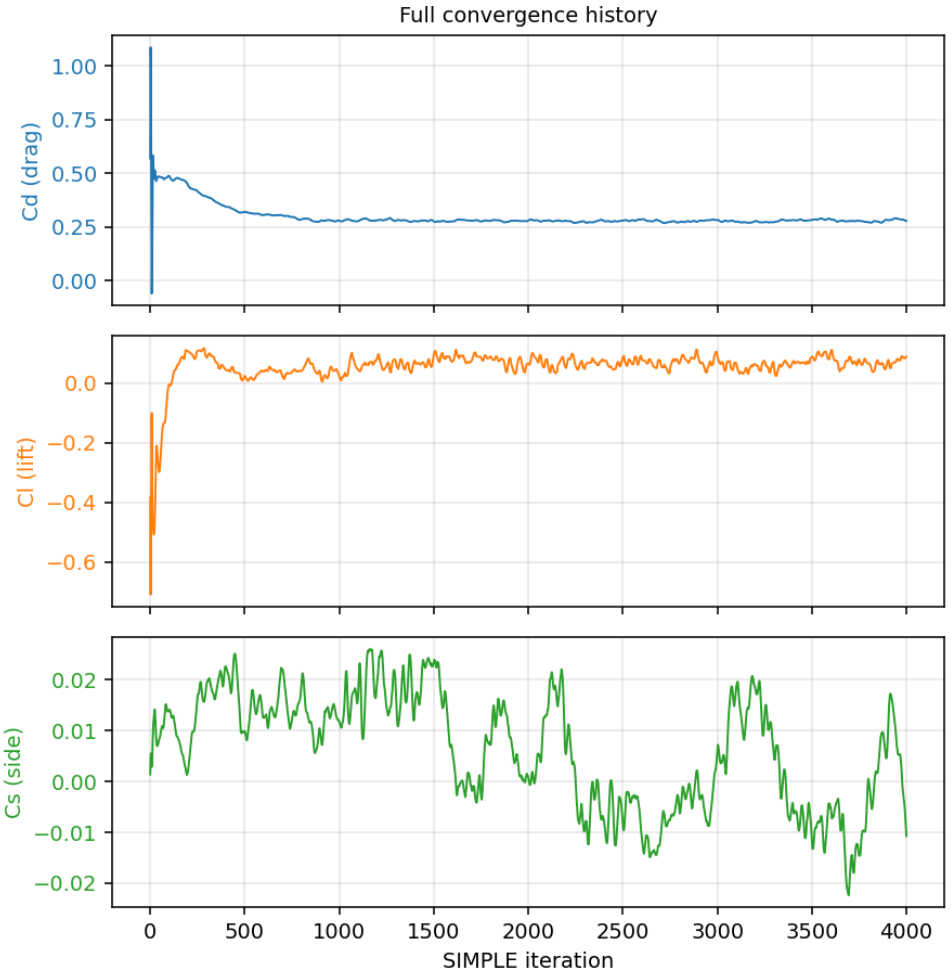
2nd OpenFOAM HPC Challenge

Gregor Olenik



+ Really challenging case, needs a lot of numerics to work correctly
+ Great for validation.
- Squeezing inherently transient case into steady state solver makes reasoning about certain optimizations harder.
- Full run is rather expensive, limits turn around for experiments.

- Track 1 Benchmark (DriveAer), Track 2 (Open Challenge)
- Prevent mindless paramter tuning



Context

Version 0.3.0 (unreleased)

Features

- Add corrected and limited-corrected face-normal gradient schemes #514
- Add experimental support for cell based assembly strategies #471,#517, #473
- Add experimental support for COO and CSR Matrices #486
- Add experimental umpire support #455
- Add experimental MPI support #519, #520, #512, #525, #531, #522
- Add python bindings via nanobind #382
- Correct the Ginkgo version to 2.0.0 (unreleased) #493
- Add tensor and symmTensor primitives, Su type sourceTerm and passing of fields to BCs #428
- Add uniform mesh generator #475
- Add distributed (processor-boundary) support: exact processor-face geometry and non-orthogonal corrected/limited snGrad correction, plus a one-time `set()` / per-iteration `update()` boundary-condition interface #528
- Add DSL expression optimizer infrastructure for operator fusing #452
- Add L1-norm stopping criterion #538

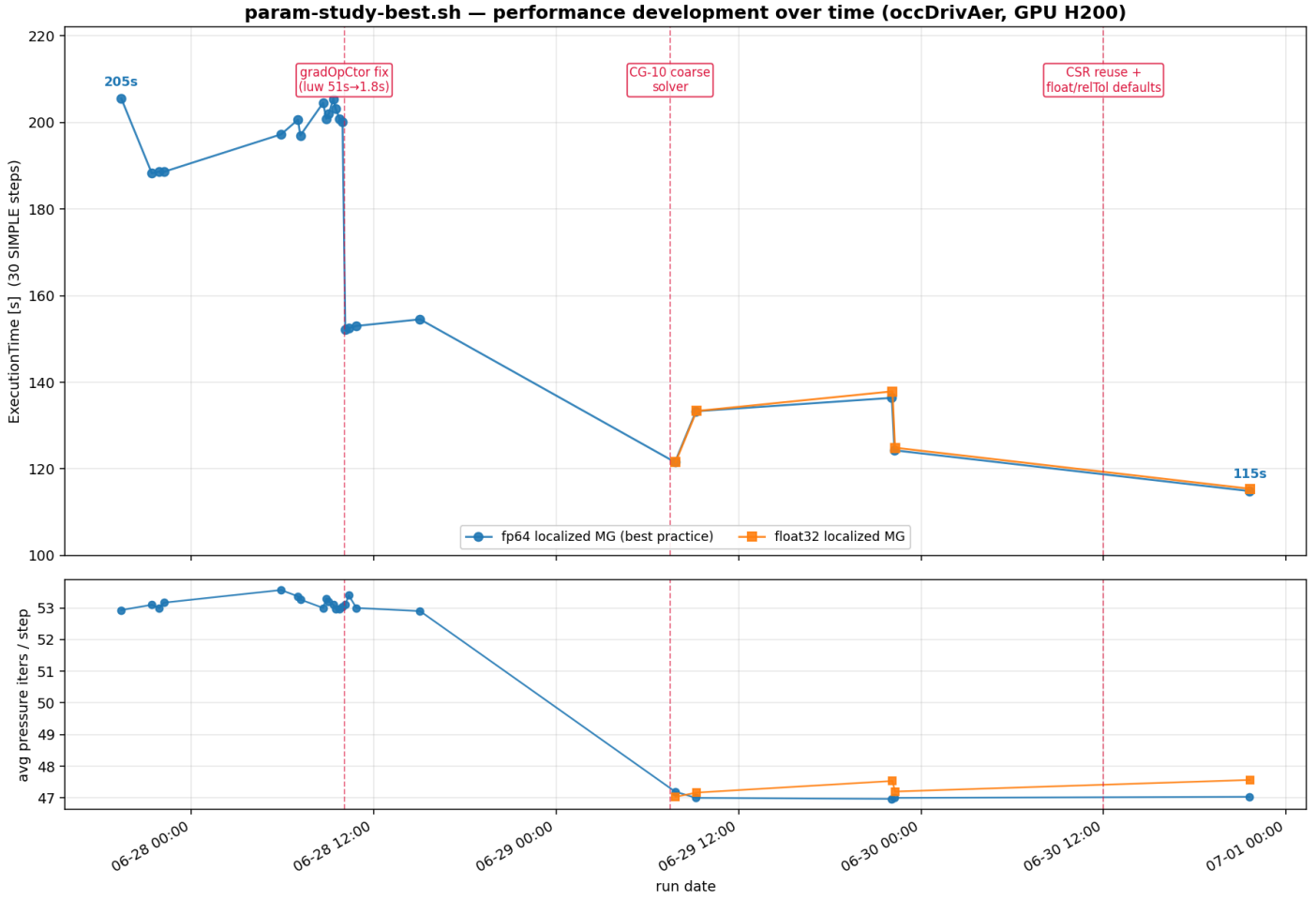
Misc

- Added linearUpwind scheme #548
- Include RHS / explicit source assembly #539

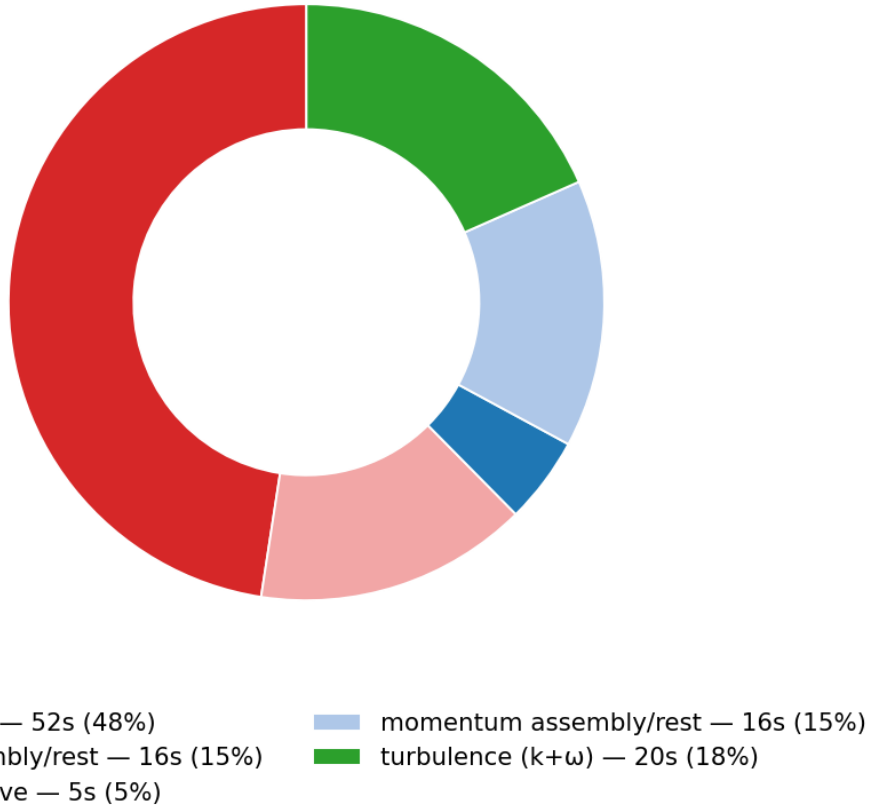
Distributed Courant number #525

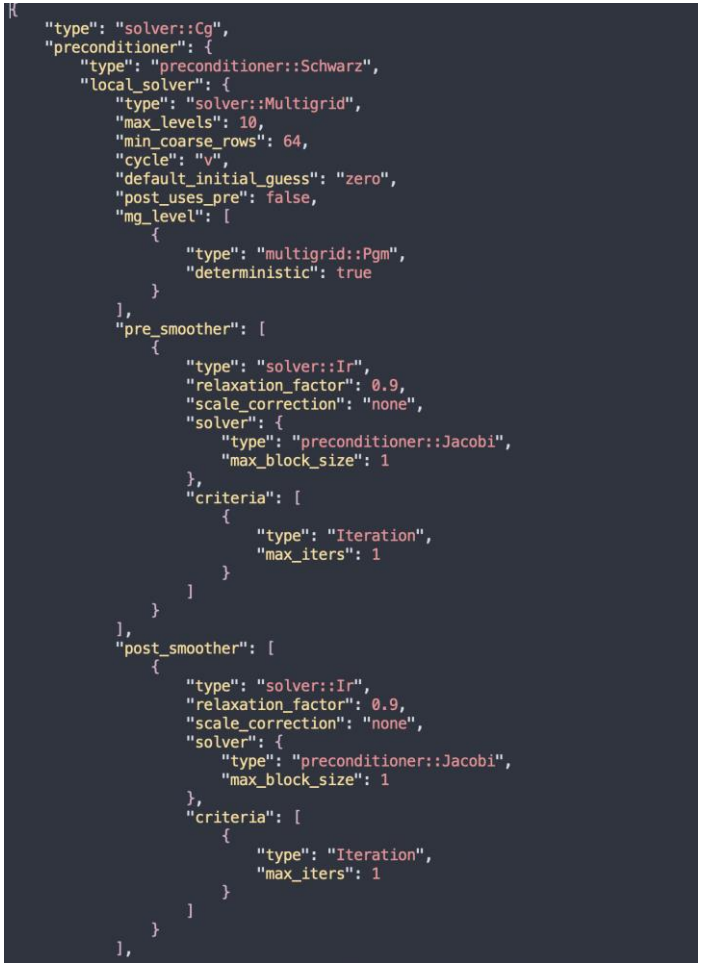
Merged greole merged 3 commits into `develop` from `enh/distributedCoNum` on May 27

Conversation 6 | Commits 3 | Checks 24 | Files changed 5


param-study-best.sh — performance development over time (occDrivAer, GPU H200)
ExecutionTime [s] (30 SIMPLE steps)
205s
gradOpCtor fix
(luw 51s→1.8s)
CG-10 coarse
solver
CSR reuse +
float/relTol defaults
115s
fp64 localized MG (best practice)
float32 localized MG
avg pressure iters / step
06-28 00:00
06-28 12:00
06-29 00:00
06-29 12:00
06-30 00:00
06-30 12:00
07-01 00:00
run date

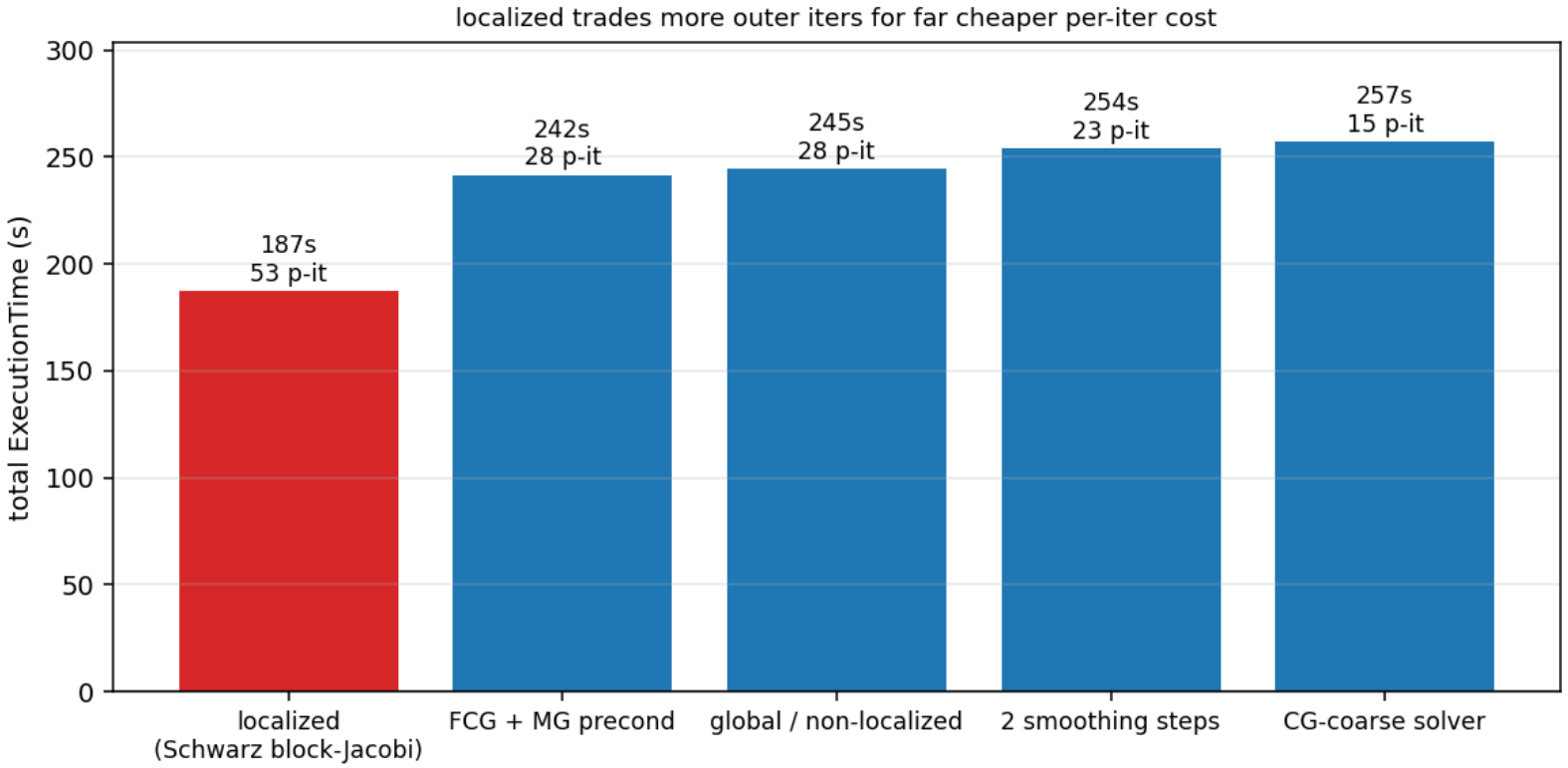
Per-step phase breakdown — 109s / 30 steps

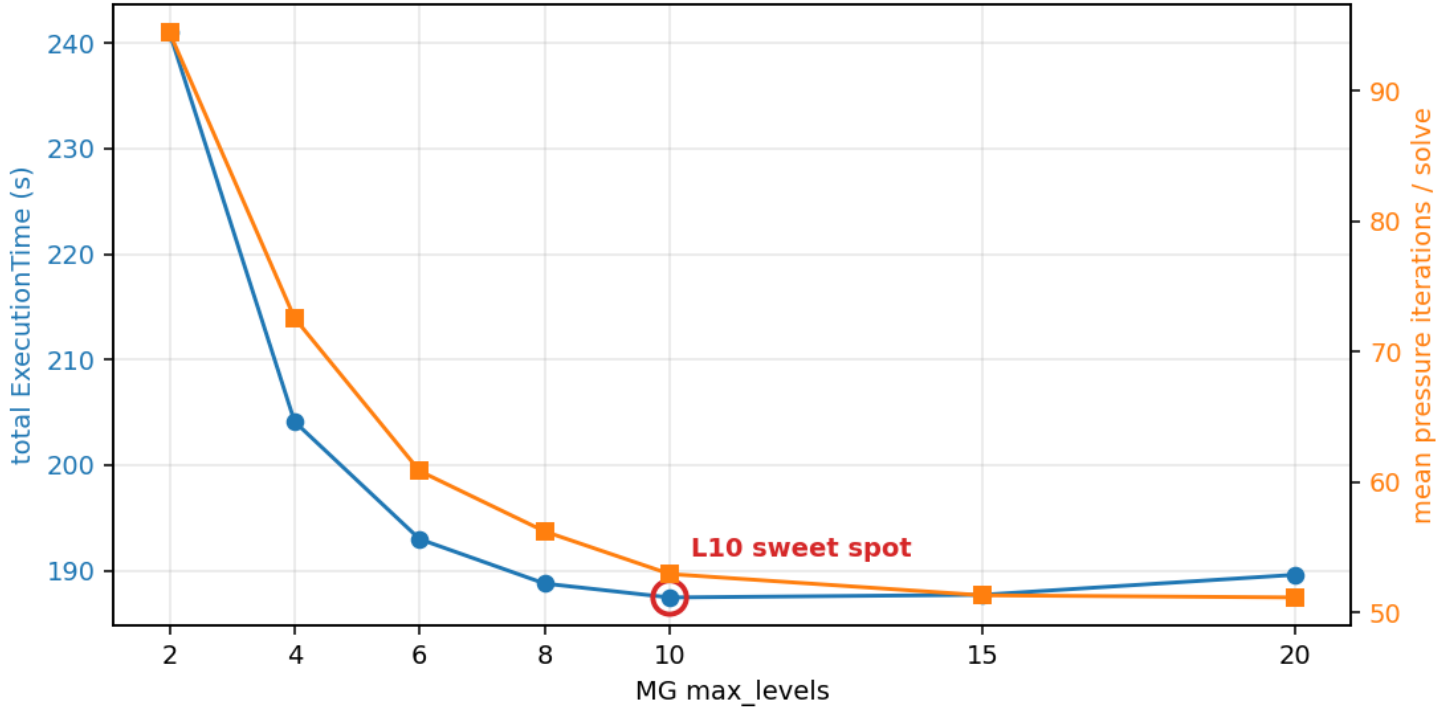




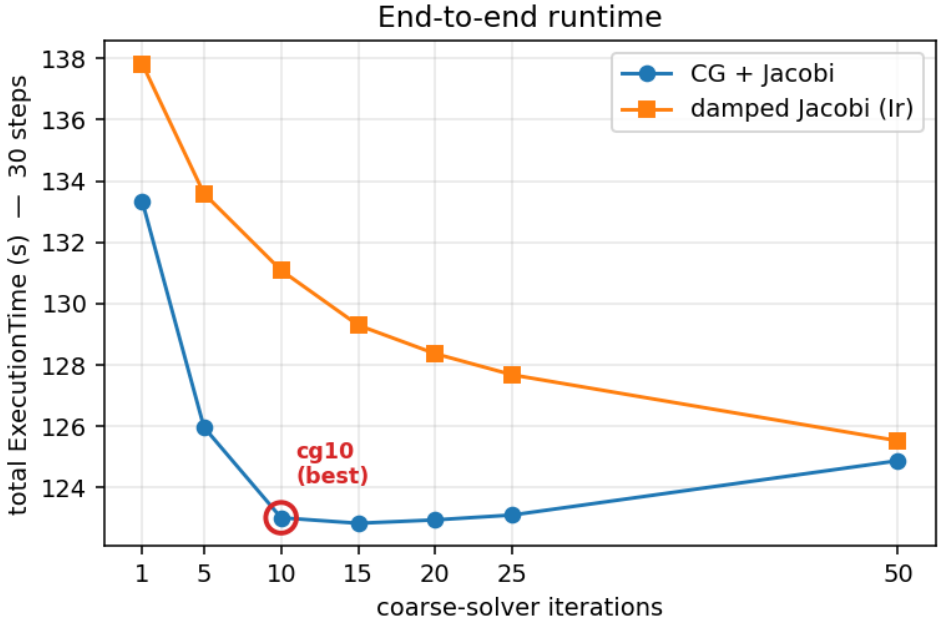
```
{
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    "preconditioner": {
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        "local_solver": {
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            "min_coarse_rows": 64,
            "cycle": "v",
            "default_initial_guess": "zero",
            "post_uses_pre": false,
            "mg_level": [
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                    "deterministic": true
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                    "relaxation_factor": 0.9,
                    "scale_correction": "none",
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                    },
                    "criteria": [
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                            "type": "Iteration",
                            "max_iters": 1
                        }
                    ]
                }
            ],
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                    "relaxation_factor": 0.9,
                    "scale_correction": "none",
                    "solver": {
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                        "max_block_size": 1
                    },
                    "criteria": [
                        {
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                            "max_iters": 1
                        }
                    ]
                }
            ],
            "coarsest_solver": {
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                "relaxation_factor": 0.9,
                "scale_correction": "forward",
                "solver": {
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                    "max_block_size": 1
                },
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                        "max_iters": 8
                    }
                ]
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                    "type": "Iteration",
                    "max_iters": 1
                }
            ]
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    },
    "criteria": [
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            "type": "Iteration",
            "max_iters": 150
        },
        {
            "type": "ResidualNorm",
            "reduction_factor": 0.01,
            "baseline": "initial_resnorm"
        }
    ]
}
```

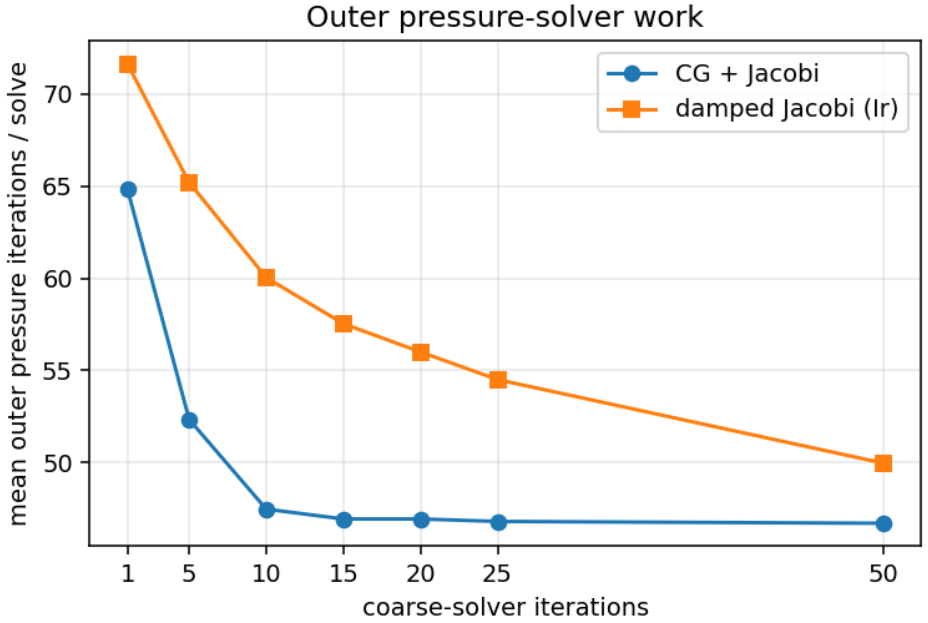
JSON config file based Ginkgo solver configuration.


localized trades more outer iters for far cheaper per-iter cost
total ExecutionTime (s)
300
250
200
150
100
50
0
187s
53 p-it
242s
28 p-it
245s
28 p-it
254s
23 p-it
257s
15 p-it
localized
(Schwarz block-Jacobi)
FCG + MG precond
global / non-localized
2 smoothing steps
CG-coarse solver



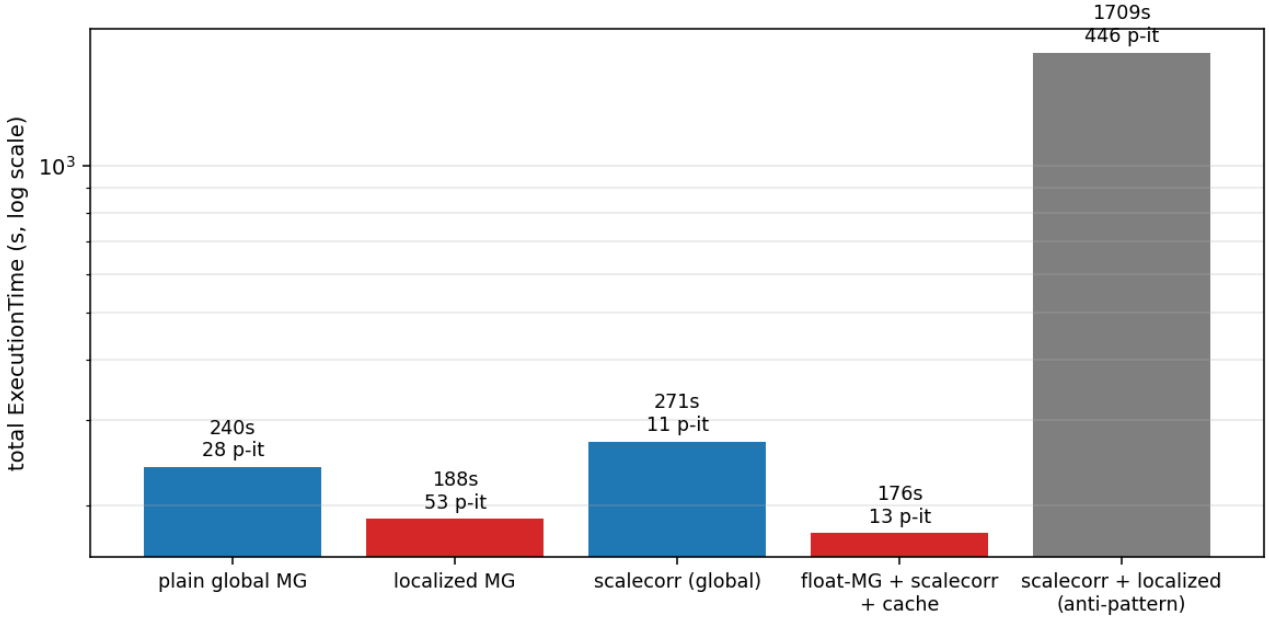
Vanilla OpenFOAM default: 50

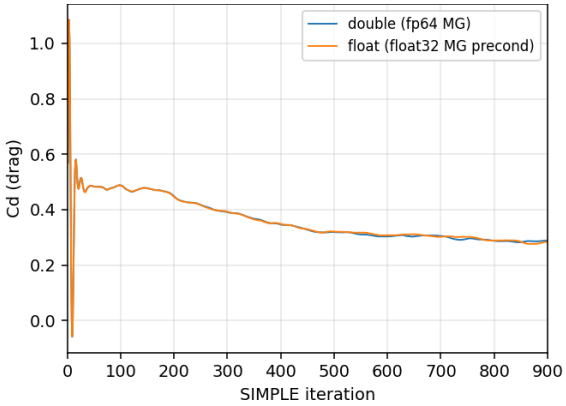

End-to-end runtime
total ExecutionTime (s) — 30 steps
138
136
134
132
130
128
126
124
CG + Jacobi
damped Jacobi (lr)
cg10
(best)
1
5
10
15
20
25
50
coarse-solver iterations

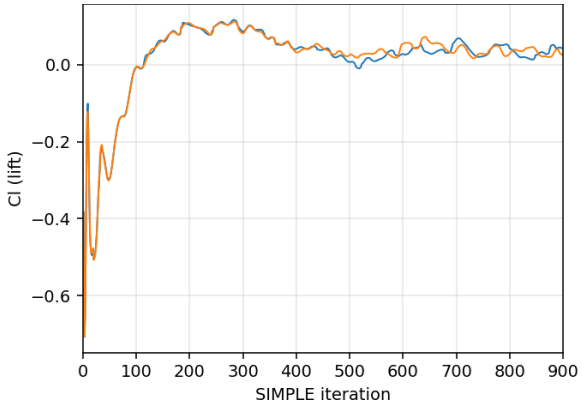

Outer pressure-solver work
mean outer pressure iterations / solve
70
65
60
55
50
CG + Jacobi
damped Jacobi (lr)
1
5
10
15
20
25
50
coarse-solver iterations

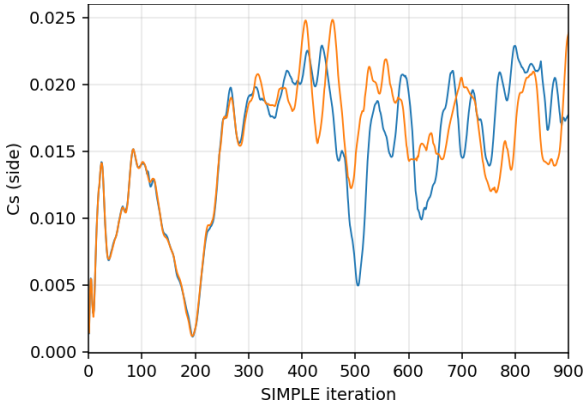
Current Shortcommings

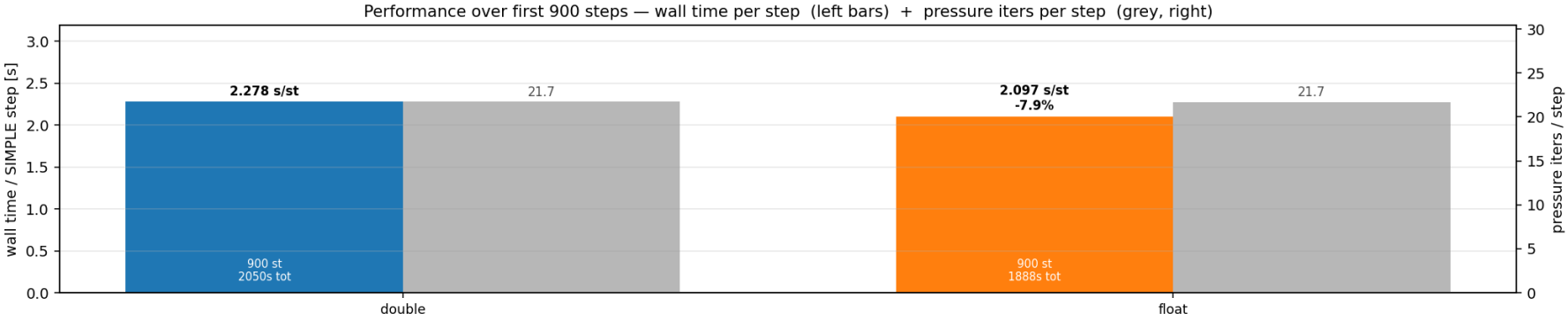
- Scale correction is known to improve convergence, currently not production ready
- Pure PGM provides limited control over coarsening. More aggressive coarsening (and less levels) potentially benefitial to reduce kernel launch overhead
- Localized + scale correction possible?

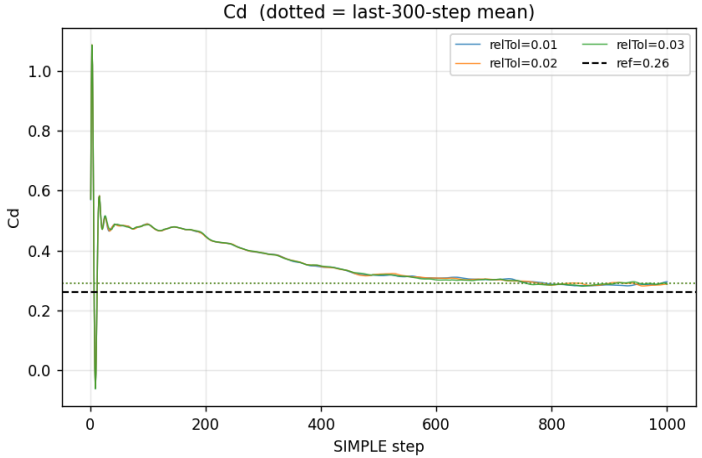


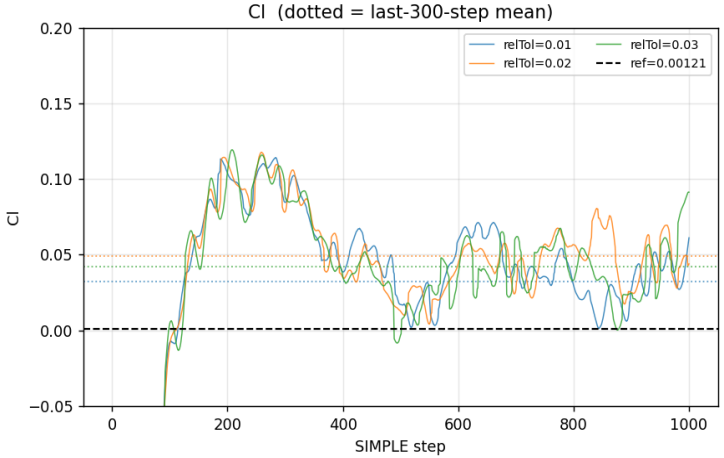

double (fp64 MG)
float (float32 MG precond)
Cd (drag)
SIMPLE iteration

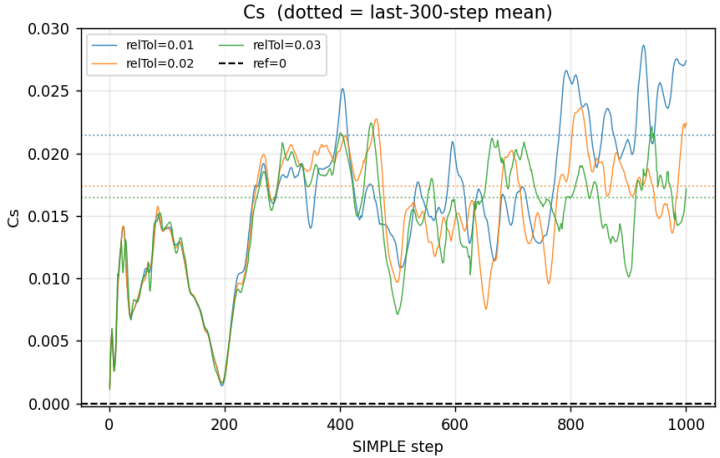

Cl (lift)
SIMPLE iteration

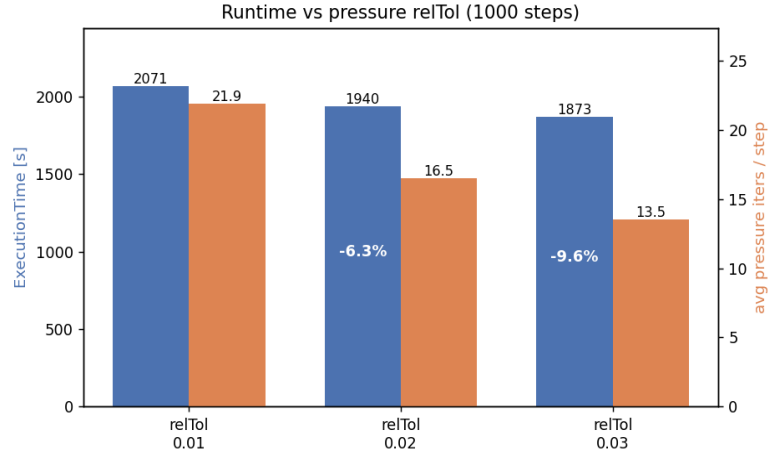

Cs (side)
SIMPLE iteration

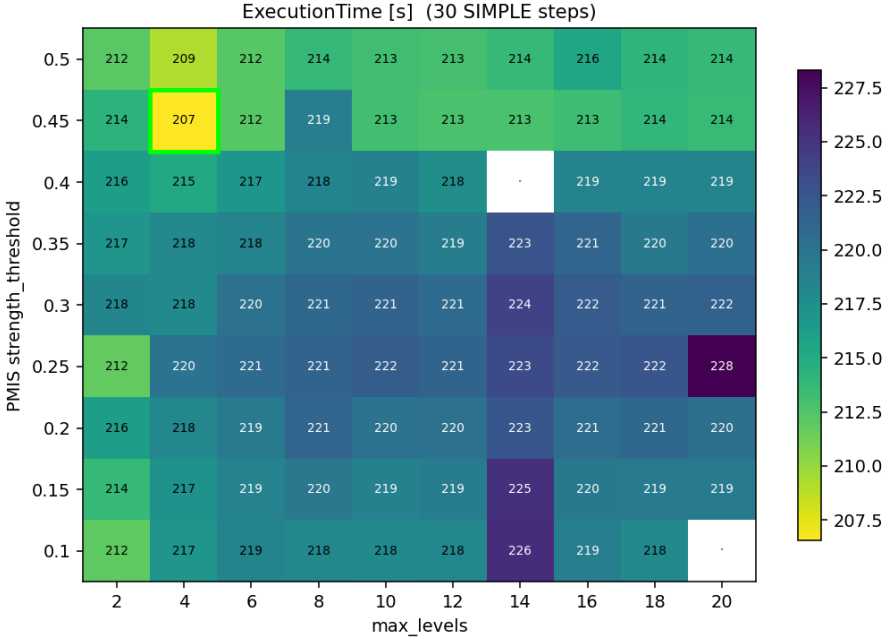

Performance over first 900 steps — wall time per step (left bars) + pressure iters per step (grey, right)
wall time / SIMPLE step [s]
pressure iters / step
2.278 s/st
21.7
2.097 s/st
-7.9%
21.7
900 st
2050s tot
900 st
1888s tot
double
float

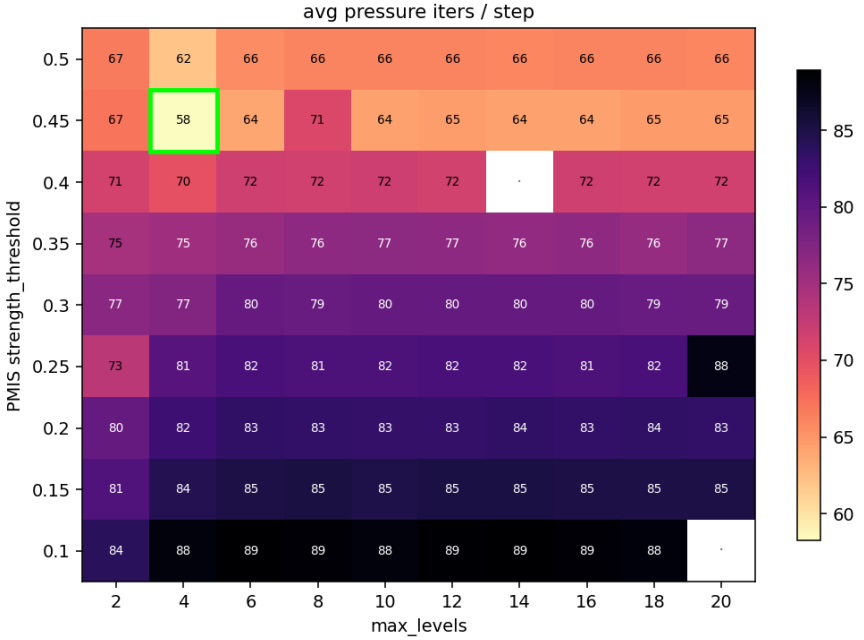

Cd (dotted = last-300-step mean)
relTol=0.01
relTol=0.02
relTol=0.03
ref=0.26
Cd
SIMPLE step

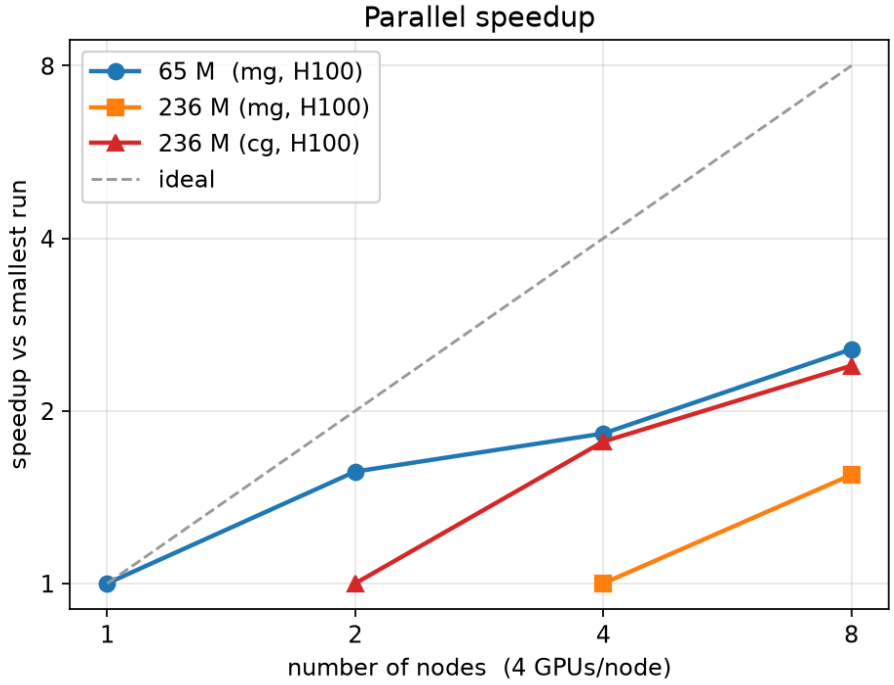

Cl (dotted = last-300-step mean)
relTol=0.01
relTol=0.02
relTol=0.03
ref=0.00121
Cl
SIMPLE step

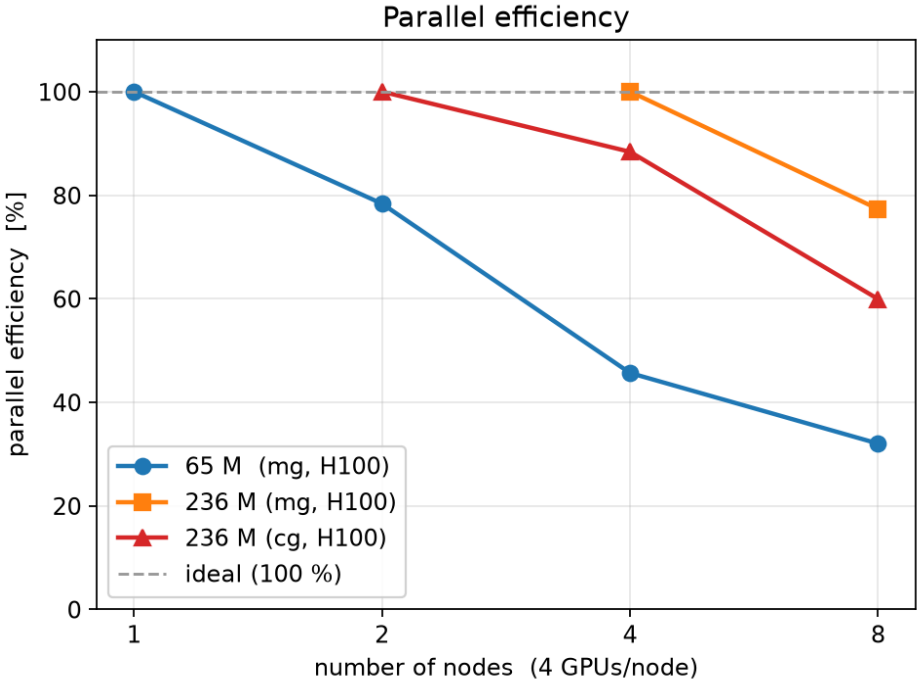

Cs (dotted = last-300-step mean)
relTol=0.01
relTol=0.02
relTol=0.03
ref=0
Cs
SIMPLE step


Runtime vs pressure relTol (1000 steps)
ExecutionTime [s]
avg pressure iters / step
2071
21.9
1940
16.5
-6.3%
1873
13.5
-9.6%
relTol
0.01
relTol
0.02
relTol
0.03


ExecutionTime [s] (30 SIMPLE steps)
PMIS strength_threshold
0.5 212 209 212 214 213 213 214 216 214 214
0.45 214 207 212 219 213 213 213 213 214 214
0.4 216 215 217 218 219 218 219 219 219
0.35 217 218 218 220 220 219 223 221 220 220
0.3 218 218 220 221 221 221 224 222 221 222
0.25 212 220 221 221 222 221 223 222 222 228
0.2 216 218 219 221 220 220 223 221 221 220
0.15 214 217 219 220 219 219 225 220 219 219
0.1 212 217 219 218 218 218 226 219 218
2 4 6 8 10 12 14 16 18 20
max_levels
227.5 225.0 222.5 220.0 217.5 215.0 212.5 210.0 207.5


avg pressure iters / step
PMIS strength_threshold
0.5 67 62 66 66 66 66 66 66 66 66
0.45 67 58 64 71 64 65 64 64 65 65
0.4 71 70 72 72 72 72 72 72 72
0.35 75 75 76 76 77 77 76 76 76 77
0.3 77 77 80 79 80 80 80 80 79 79
0.25 73 81 82 81 82 82 82 81 82 88
0.2 80 82 83 83 83 83 84 83 84 83
0.15 81 84 85 85 85 85 85 85 85 85
0.1 84 88 89 89 88 89 89 89 88
2 4 6 8 10 12 14 16 18 20
max_levels
85 80 75 70 65 60


Parallel speedup
65 M (mg, H100)
236 M (mg, H100)
236 M (cg, H100)
ideal
speedup vs smallest run
number of nodes (4 GPUs/node)
1
2
4
8


Parallel efficiency
65 M (mg, H100)
236 M (mg, H100)
236 M (cg, H100)
ideal (100 %)
parallel efficiency [%]
number of nodes (4 GPUs/node)
0
20
40
60
80
100
1
2
4
8